

Proving **the engine**

How we verify each phase ships green, existing tenants are untouched, and the card only ever moves through one audited path.

OWNER
Engineering / QA

COVERS
Model · Engine · Migration · UI

BASELINE
553 tests stay green

GATE
Per-phase exit checklist

OBJECTIVES

Four things must be true when this ships

01

One path moves the card

workflows.stage is only ever written by the engine — never a raw UI drag or a stray endpoint.

02

Backward compatible

Empty stage templates reproduce today's behaviour exactly; no tenant sees a change until they opt in.

03

Migrations are safe

Exactly-once, idempotent, replica-restart-safe. Correct join keys, no data lost.

04

One shared lifecycle

Closing the last stage task advances the card; the card advancing spawns the next stage's tasks.

STRATEGY

Where each behaviour is proven

LEVEL

WHAT IT COVERS

Unit

on_stage_enter · check_stage_ready · advance; stage-contract evaluation; task-spawn from templates

Integration

Task-closer → engine → transition → next-task spawn; approval recording → gate release

Migration

Backfill correctness, idempotency, replica-restart replay, rollback safety

Contract / API

Only the engine writes stage; manual advance requires override + reason

UI / E2E

Settings Operations tab, stage chips, inline task lists, manual-advance reason prompt

Regression

Full existing suite (553) + ad-hoc tasks (null workflow_id) still work

Unify the model — behaviour unchanged

MIGRATION / DATA

- **Backfill sets workflow_id + stage_key**
correctly, via the decision_id join
- **No-match tasks keep null fields**
and still load, edit, and close
- **Re-running the backfill changes nothing**
idempotent

BEHAVIOUR & UI

- **Voice capture creates the same card + tasks**
My Work, Workflows, counters unchanged
- **Ghost collections gone**
no reader errors on their absence
- **Settings renders as four tabs**
every old field reachable

EXIT GATE

Full suite green · zero behaviour diff · both ghost collections removed.

The engine — three operations under test

OPERATION

WHAT THE TESTS ASSERT

on_stage_enter

Spawns exactly the template tasks for the named roles · records timeline · notifies · empty template spawns nothing

check_stage_ready

False while any task open · false while any required approval missing · true only when all done AND all present

advance

Pass → transition → chained on_stage_enter · terminal → side-effects + done · fail → no change, clear reason

PHASE 2

One path, always audited

- ✓ Single writer — no path other than the engine mutates workflows.stage (contract test scans writers)
- ✓ Manual advance routes through the engine with override=true + a required reason, written to audit
- ✓ Idempotent transitions — calling advance twice never double-spawns or skips a stage
- ✓ Concurrency — two tasks closing at once trigger exactly one transition (no double-advance)
- ✓ Side-effects fire once — procurement-terminal → awaiting-bill creates exactly one expense

Flat strings → stage objects, non-destructively

- **Each old string → {key, label, tasks:[], approval:null}**
with a stable, unique key; label preserves the display string; order preserved
- **Empty tasks / approval → engine spawns nothing**
gate passes trivially — today's behaviour preserved until owners edit templates
- **In-flight cards keep their current stage**
and advance correctly afterward
- **Idempotent + replica-restart-safe via the ledger**
dry-run diff reviewed before apply

The joined view — two lenses, one truth

MY WORK

- **Every task shows a stage chip**
"Order #4821 · Confirmed" — click opens the parent card
- **Ad-hoc tasks show no chip**
and behave exactly as before

WORKFLOW BOARD

- **Each card lists its current stage's open tasks**
inline, with assignee avatars
- **Closing an inline task updates both views**
card + My Work, no manual refresh

MANUAL ADVANCE

Prompts for a reason · empty reason rejected · reason stored in history · retired Board sub-tab is gone.

The golden-path test — the Kapoor order

- 1 Voice capture → card at Booked + task "Confirm with customer" (Sales)
- 2 Sales closes task → advance to Confirmed → "Prepare invoice" (Finance) + Owner approval pending
- 3 Finance closes task AND Owner approves → advance to Dispatched → "Pack & ship" (Ops)
- 4 Ops closes task → advance to Paid (terminal) → side-effect marks invoice paid → card done
- 5 History shows every transition — no manual advance was ever needed

Existing tenants must not feel this

553

baseline tests stay green at every phase boundary

- Tenants who never edit templates see identical behaviour
- Ad-hoc tasks (no workflow) create, assign, close as today
- Counters, dashboards, notifications unchanged until a template is populated
- Each phase has a tested reverse or forward-fix — app stays shippable

GO / NO-GO

A phase isn't done until its gate is green

PHASE

EXIT GATE

Phase 1

Ghosts removed · backfill idempotent · zero behaviour diff · 4-tab Settings verified · suite green

Phase 2

Engine unit + integration green · single-writer contract holds · stage migration replica-safe · side-effects fire once

Phase 3

Stage chips + inline tasks verified · manual advance needs a reason · Board sub-tab retired · E2E golden path green

NOT TESTED NOW

Deliberately excluded — **matches the build's non-goals**

Cross-pipeline dependencies

SLA timers

Workload balancing

Forecast reports

We test that the board can be trusted again — not features we haven't built yet.

Ship the engine, watch it run, then scope the next test cycle.